

Project Specification: Deep Investigation and Custom Design of Spatial Filters for Image Enhancement

Course: CSE 4106 - Image Processing (CO2, CO3 Focus)

Project Title: In-Depth Investigation and Performance Comparison of Standard vs. Custom-Designed Spatial Filters for Image Enhancement

Total Marks: 25 + 10 (Presentation + Report)

Due Dates:

- **Code & Report Submission:** [TBA]
- **Project Presentation:** [TBA]

1. Project Objective (Assessing K1, K2, P1, P3, P5)

This project aims for a rigorous, in-depth investigation into spatial domain filtering, culminating in the design and critical evaluation of custom filters. Students will:

- **Investigate** the core principles and mathematical foundations behind different classes of spatial filters (linear/non-linear smoothing, derivative-based sharpening).
- **Implement at least four standard spatial filters** from the curriculum.
- **Design and implement at least two novel or custom spatial filters** (one for smoothing, one for sharpening) tailored to specific degradation scenarios or desired enhancement effects.
- **Utilize modern image processing tools and environments** for robust implementation, experimental design, and comprehensive analysis.
- **Critically compare and evaluate** the performance of their custom filters against existing standard filters, focusing on effectiveness in noise reduction, edge preservation, detail enhancement, and computational efficiency. This comparison will drive a deeper understanding of filter design principles.

Mapping:

- **K1, K2, K3, K4 (Problem Solving, Theory, Design):** Deep application of theoretical knowledge to design and analysis.
- **P1, P3, P5 (Structured Experimentation, Judgment & Evaluation):** Systematic investigation and critical judgment in filter design and comparison.
- **CO2, CO3:** Direct application and evaluation of enhancement/restoration techniques.

2. Scope & Methodology

Students will comprehensively investigate spatial filtering by:

1. **Implementing at least four standard spatial filtering techniques** (as per the curriculum breakdown: at least two smoothing, two sharpening).
 - **Smoothing:** At least one linear (Box, Weighted Average) and one non-linear (Median, Min, Max).
 - **Sharpening:** At least one first-order (Sobel) and one second-order (Laplacian, Unsharp Masking, High boost).
2. **Designing and implementing at least two custom spatial filters:** [See Note below]

- **Custom Smoothing Filter:** Students will design a filter kernel or a non-linear operation (e.g., a Gaussian-weighted median, an adaptive mean filter based on local variance, a specialized edge-preserving smoothing filter).
 - **Custom Sharpening Filter:** Students will design a new derivative kernel, a hybrid derivative operator, or a modified Unsharp Masking/High boost variant with specific properties (e.g., anisotropic sharpening, emphasizing specific orientations, or a more subtle high-boost effect).
3. **Acquiring or simulating at least three diverse grayscale images** for robust testing:
- Image 1: With significant noise (e.g., Gaussian, Salt-and-Pepper) and smooth regions.
 - Image 2: With fine details and sharp edges.
 - Image 3: A complex scene combining both noisy areas and important details.
 - **Degrade** these images appropriately (add noise, induce blur) to create challenging test cases for both smoothing and sharpening.
4. **Conducting a structured experimental comparison:**
- Apply all standard and custom filters to the degraded test images.
 - **Evaluate** results both quantitatively (PSNR/SSIM for smoothing, potential edge strength metrics for sharpening, computational time) and qualitatively (visual inspection, artifact generation, detail preservation).
 - **Crucially, compare their custom filters' performance against relevant standard filters** for similar tasks.

3. Deliverables and PO Mapping

The project will be assessed through the following deliverables, explicitly designed to map to the target POs, Ks, Ps, and As.

3.1. Project Report (Primary for PO5, Supports K1-K4, K6, K8, P1, P3, P5, A1, A3)

The report should be a comprehensive document (e.g., 15-20 pages, excluding appendices) that acts as the core technical documentation.

1. Introduction & Problem Statement (K1, K2, P1):

- Clearly define the role of spatial filtering in image enhancement and restoration.
- Distinguish between smoothing and sharpening, and the need for optimized or custom solutions.
- State the project's specific objectives, emphasizing the investigation and custom design components.

2. Theoretical Background & Mathematical Models (K2, K3, K4, P1):

- For **each implemented standard filter:** Explain its principles, kernel, and operation.
- For **each custom-designed filter:**
 - **Justify the design choice:** Explain *why* this specific custom filter was designed, referencing the limitations of standard filters or a specific desired effect.
 - Present its mathematical model, kernel (if applicable), and detailed operational steps.

- Discuss the theoretical rationale behind its expected performance.
- Reference relevant literature.

3. Methodology & Implementation Details (K3, K4, K6, P3, PO5):

- **Modern Tools Usage (PO5):** Clearly state programming language (Python/MATLAB), specific libraries (NumPy, OpenCV, SciPy.signal), and justify their selection for efficient implementation and custom filter development.
- **Algorithm Steps:** Detail implementation for all standard and custom filters.
- **Test Images & Degradation:** Describe chosen test images and how they were degraded (e.g., specific noise parameters, blur kernels).
- **Parameter Tuning:** Discuss parameter selection for both standard and custom filters.

4. Experimental Results & Evaluation (K3, K6, K8, P3, P5, A3):

- **Visual Results:** Present original, degraded, and all filtered images (standard and custom) for each test image. Use clear figures and captions, highlighting effects.
- **Quantitative Metrics (K8, P5):**
 - For smoothing: PSNR, SSIM, MSE (against original).
 - For sharpening: Consider methods to quantify edge preservation/enhancement (e.g., comparing edge maps from Canny, or subjective visual ranking if objective metrics are difficult). Computational time for each filter.
- **Analysis of Results:** Discuss how each filter performs with different image content and degradation types.
- **Custom Filter Performance:** Explicitly analyze how the custom filters compare to relevant standard filters in terms of both quantitative metrics and visual impact.

5. Discussion & Comprehensive Comparison (K6, K8, P5, A3):

- **Deep Comparative Analysis:** Critically compare all implemented spatial filters, with a strong focus on **how the custom filters address gaps or excel in specific areas where standard filters fall short.**
- **Strengths & Weaknesses (P5):** Detail the contexts where each filter (standard and custom) excels or is limited.
- **Trade-offs (P5):** Discuss the inherent trade-offs between noise suppression, detail preservation, edge enhancement, and computational complexity, specifically highlighting how custom designs attempt to optimize these.
- **Real-world Implications:** Conclude the practical applicability of each technique, especially for their custom designs, for different image types or specific enhancement goals.

6. Conclusion & Future Work (K8, P5): Summarize key findings, the success of custom designs, and suggest potential improvements or future research directions (e.g., adaptive filtering, combining custom smoothing/sharpening).

7. References (K1): Proper academic citation of all sources.

Appendices: Full source code with comments.

3.2. Project Presentation (Primary for PO4, Supports K1-K4, K6, K8, P1, P3, P5, A1, A2, A4)

The presentation (12-15 minutes, followed by 3-5 minutes Q&A) will serve as the primary assessment for **PO4 (Investigation)**. Students must demonstrate their in-depth investigative process and design rationale.

1. Introduction & Problem (K1, P1, A1): Clearly state the problem and project objectives, emphasizing the need for investigation and custom filter design.

2. Spatial Filtering Overview & Identified Gaps (K2, K3, A1): Briefly explain smoothing/sharpening and *critically discuss why standard filters might be insufficient or suboptimal* for certain scenarios, setting the stage for custom designs.

3. Investigative Journey & Custom Filter Design (PO4, K4, K6, P3, A2):

- **This section is paramount for PO4.** Students must articulate *how* they investigated the problem and *why* their custom filters were designed.
- **Rationale for Custom Design:** Explain the specific limitations of standard filters that their custom design aims to overcome.
- **Design Process:** Walk through the thought process for designing each custom filter. Show iterations, discarded ideas, and the final chosen approach. Explain the theoretical basis for their custom kernel/operation.
- **Implementation Challenges:** Discuss challenges encountered in implementing both standard and custom filters, and their solutions.
- **Parameter Exploration:** Show how varying parameters (for both standard and custom filters) impacted results, demonstrating methodical exploration.

4. Demonstration & Results (K3, K8, P5, A2):

- Present original, degraded, and all filtered images (standard and custom).
- Clearly display relevant quantitative metrics and explain their significance.
- **Visually emphasize where custom filters perform better or offer a unique effect compared to standard ones.**

5. Critical Analysis & Comprehensive Discussion (PO4, K6, K8, P5, A4):

- **Deep Comparative Analysis:** Directly compare the performance of all filters.
- **Custom Filter Validation:** Critically assess whether the custom filters achieved their intended design goals and whether they truly outperform or offer a distinct advantage over standard filters in specific contexts.
- **Trade-off Analysis:** Emphasize the trade-offs, particularly how custom designs navigated these.
- **Limitations of Custom Designs:** Acknowledge the limitations or potential weaknesses of their *own custom designs*.
- **Future Investigation:** What *further investigation* is needed based on their findings?

6. Q&A (PO4, A4): Engage in a detailed Q&A session, demonstrating comprehensive understanding, ability to defend design choices, and discussing broader implications.

4. K, P, A Mapping (from Table H)

- **Knowledge Profile (K):**
 - **K1 (Math, Science):** Applied in understanding convolution, derivatives, statistical filter properties, and in **mathematically deriving/designing custom filter kernels/operations.**
 - **K2 (Discipline-specific):** Core image processing concepts of spatial domain filtering, noise models, edge detection theory, and **applying these to custom filter design.**
 - **K3 (Problem-Solving):** Selecting appropriate filters and **designing novel solutions for specific problems.**
 - **K4 (Engineering Sciences):** Understanding input-output behavior and **predicting and analyzing the behavior of custom-designed filters.**
 - **K6 (Design/Development):** Designing experimental setup, implementing various filters, and **critically designing and developing custom algorithms.**
 - **K8 (Contemporary Issues):** Understanding the role of spatial filtering and **the need for specialized solutions in advanced applications.**
- **Problem Level (P):**
 - **P1 (Broadly defined):** Understanding the real-world need for clearer, sharper images.
 - **P3 (Structured Experimentation):** Designing, conducting, and analyzing controlled experiments with standard and **custom-designed** filters.
 - **P5 (Judgment & Evaluation):** Critically evaluating performance, drawing conclusions, and **making informed judgments about filter design effectiveness.**
- **Assessment (A):**
 - **A1 (Proposal/Plan):** Implicit in the project set up and initial understanding.
 - **A2 (Demo/Application):** Through implementation and showing filtered images and **demonstrating the functionality of custom designs.**
 - **A3 (Report):** Comprehensive written documentation, including **detailed design rationale for custom filters.**
 - **A4 (Presentation):** Oral communication of findings, defending choices, engaging in discussion, AND **articulately presenting the investigative and design process.**

5. Marking Scheme (25 + 10 Marks)

- **Problem Identification & Objective Definition (3 + 1 Marks):** Clarity, relevance, and distinction between standard/custom needs.
- **Theoretical Background & Custom Design Rationale (4 + 2 Marks):** Accuracy, depth of explanation for all filters; **clear justification and mathematical/algorithmic detail for custom designs.**
- **Implementation (6 + 2 Marks):** Correctness, efficiency, and clarity of code for *all* filters (standard and custom); **effective use of modern tools for custom filter development (PO5).**

- **Experimental Results & Evaluation (6 + 2 Marks):** Presentation of visual and quantitative results; **insightful analysis of custom filter performance against standard filters** (CO2, CO3).
- **Discussion & Comprehensive Comparison (4 + 2 Marks):** Critical comparison across *all* filters; **in-depth analysis of custom filter strengths/weaknesses and design decisions; demonstration of investigative depth (PO4).**
- **Report & Presentation (2 + 1 Marks):** Overall clarity, structure, visuals, and oral delivery; **compelling communication of investigative process and custom design journey (PO4), and tool usage (PO5).**

[Note]

Clarification: What Constitutes a "Custom Spatial Filter"?

It does not necessarily mean inventing an entirely new filter never seen before (though that would be impressive!). It means demonstrating thoughtful modification or combination.

Examples of Custom Smoothing Filters:

The goal here is typically to improve noise reduction while better preserving edges or fine details, or to target a specific noise type.

1. Weighted Median Filter:

- **Concept:** Instead of all neighbors having equal "vote" in a median filter, some pixels (e.g., the center pixels, or pixels closer to the center) are given higher weights (i.e., they appear multiple times in the sorted list before finding the median).
- **Customization:** Students define a custom weighting mask for median operation.
- **Rationale:** To make the median filter less aggressive on fine details while still effective against impulse noise.

2. Adaptive Mean/Gaussian Filter:

- **Concept:** The filter's properties (e.g., kernel size, standard deviation for Gaussian, or even its shape) change based on local image characteristics.
- **Customization:** Design a mean or Gaussian filter where the filter size or parameters are adjusted based on the local variance of the image patch. For instance, smaller kernels in high-variance (edge) regions to preserve edges, larger kernels in low-variance (smooth) regions for more aggressive noise reduction.
- **Rationale:** To achieve better edge preservation than a global mean/Gaussian filter.

3. Directional Smoothing Filter:

- **Concept:** Instead of a square kernel, the filter averages pixels only along a specific orientation (e.g., horizontal, vertical, or diagonal line segments) where an edge might be present.
- **Customization:** Implement several small, elongated kernels, apply them, and then choose the result from the kernel that aligns best with the local image structure (e.g., lowest local variance).
- **Rationale:** To smooth noise along edges without blurring the edge itself, which a square kernel would do.

4. Hybrid Filter for Specific Noise:

- **Concept:** Combine elements of different filters.
- **Customization:** For example, a filter that first applies to a small median filter (good for salt-and-pepper) and then a light Gaussian filter (good for remaining Gaussian noise), or a filter that detects high-frequency spikes (noise) and selectively applies a median only to those pixels.
- **Rationale:** To address mixed noise types more effectively than a single filter.

Examples of Custom Sharpening Filters:

The goal here is typically to enhance edges more selectively, reduce noise amplification during sharpening, or target specific features.

1. **Custom Laplacian Kernel:**

- **Concept:** Modify the standard Laplacian kernel weights.
- **Customization:** Design a 3X3 or 5X5 Laplacian-like kernel where the central weight and/or surrounding weights are adjusted. For instance, giving higher weights to diagonal elements, or making it less aggressive.
- **Rationale:** To emphasize edges in certain orientations, or to produce a softer/harder sharpening effect than standard Laplacians. (Ensuring the sum of weights is 0 for a pure derivative kernel).

2. **Directional Gradient Operator:**

- **Concept:** Standard Sobel/Prewitt operators are typically for horizontal and vertical gradients.
- **Customization:** Design kernels to detect gradients along diagonal directions (e.g., 45 degree or 135 degree or to combine gradient information in a custom way beyond simple magnitude (e.g., emphasizing a specific direction)).
- **Rationale:** To enhance edges in specific orientations that might be missed or attenuated by standard operators.

3. **Adaptive Unsharp Masking/High boost Filtering:**

- **Concept:** Unsharp Masking relies on subtracting a blurred version of the image.
- **Customization:** Make the parameters (e.g., the blurring filter type/strength for the 'unsharp' part, or the gain factor for High boost) adaptive to local image features (e.g., higher gain in areas with weak edges, lower gain in already sharp or noisy areas).
- **Rationale:** To sharpen more aggressively where needed, but avoid over-sharpening or noise amplification in other regions.

4. **Pre-filtered Sharpening:**

- **Concept:** Sharpening amplifies noise.
- **Customization:** Implement a multi-stage approach where the image is first lightly smoothed (e.g., with a small median filter) *before* applying a sharpening filter, or combining a sharpening filter with a noise detector to apply sharpening selectively.
- **Rationale:** To achieve sharpening while actively mitigating noise amplification.